STATISTICS 321: Practice Lab

Purpose

This lab is designed to prepare you for upcoming quizzes and examinations by practicing the types of reasoning expected in this course. You may use AI tools, notes, textbooks, class handouts, R, and other course resources. The goal is not simply to obtain answers, but to understand what is being expected, why a method works, when a method is appropriate, when a method becomes inefficient, and what assumptions are required.

Expectations

Show reasoning clearly. Use probability notation appropriately. Support claims with mathematical arguments. Explain why methods work and when they apply. If you disagree with a solution, identify the first step where the reasoning fails.

Question 1: Choosing and Justifying a Strategy

For each scenario: (a) identify the approach you would recommend first; (b) write the first counting or probability expression you would begin with; (c) identify a second approach that would also produce a correct answer; (d) explain why your preferred approach is more efficient.

Available approaches: Explicit sample space enumeration, Outcome table, Complement rule, Combination counting, Union/intersection reasoning, Sequential multiplication, Symmetry argument.

- **A.** Fair coin flipped 15 times. Determine $P(\text{at least one Head})$.
- **B.** A player wins any game with probability 0.5. Six games are played. Let X be the number of wins. Determine $P(X = 4)$.
- **C.** One card is drawn from a standard 52-card deck. Let $K =$ King and $H =$ Heart. Determine $P(K \cup H)$.
- **D.** Two cards are drawn without replacement. Determine $P(\text{two Aces})$.
- **E.** Two fair six-sided dice are rolled. Determine $P(X_1 > X_2)$.

Follow-Up: Choose two scenarios. For each: (i) show how your alternate method would begin; (ii) explain why the alternate method still works mathematically; (iii) explain what feature makes the alternate method cumbersome; (iv) describe one small change that would make your preferred method less attractive.

Question 2: Comparing Three Solutions

A fair six-sided die is rolled three times. Let X = number of sixes observed. Determine $P(X = 2)$.

Student A: Uses arrangements $(6, 6, 6^c)$, $(6, 6^c, 6)$, $(6^c, 6, 6)$; counts $\binom{3}{1} = \frac{3!}{1!2!} =_3 C_1 = C_1^3 = 3$ locations for the non-six and computes

$$3\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right).$$

Student B: Uses $n(S) = 6^3 = 216$ equally likely outcomes. Counts 15 favourable outcomes and computes

$$\frac{n(A)}{n(S)} = \frac{15}{216}.$$

Student C: Claims

$$P(X = 2) = 3\left(\frac{1}{6}\right)^2 = \frac{1}{12}$$

because there are three arrangements containing two sixes.

Answer:

(a) Verify algebraically that Students A and B obtain the same probability.

(b) Explain precisely what $\binom{3}{1}$ counts.

(c) Identify where the same counting idea appears in Student B's solution.

(d) Explain the difference between arrangements and outcomes.

(e) Evaluate the claim that Student B is more correct because actual outcomes were counted.

(f) Determine whether Student C is correct.

(g) Identify the first incorrect step.

(h) Explain why Student C's answer differs from the correct answer.

(i) Rewrite Student C's solution correctly.

(j) Generalize to a 20-sided die rolled 15 times and $P(X = 3)$. Which strategy would you recommend?

(k) Write the first expression produced by that strategy.

(l) Does preferring one method imply the other is mathematically incorrect?